

Econ 131
Spring 2026
Danny Yagan

Problem Set 3

DUE: May 1, 2026, at 5PM PT on Gradescope

The problem set is due on Gradescope at 5pm sharp on the due date, and solutions will be posted at noon the next day or shortly thereafter. If you submit it to Gradescope between the 5pm deadline and the noon posting, you can get up to half credit. (If you have a DSP accommodation for extra time on take-home assignments, you will get up to full credit if you submit it by the noon deadline.) No submissions after the noon deadline will be accepted for any student, regardless of when solutions are posted.

Student Name:

Student ID:

- Write or type your answers clearly and in dark ink (physical or electronic ink) so that your responses are legible.
- Tag each of your answers on Gradescope so that it is clear what responses are to which questions.
- **Although you may work in groups**, each student must submit individual sets of solutions. You must note the names of other students that you worked with. Write their names here:

1. True/False/Uncertain with Explanation [30 points]

Determine whether each statement is true, false, or uncertain and explain why. Answers with no explanation will receive no points. [5 points each]

- (a) If schooling is a screening device, the government should encourage more education.

- (b) In the standard expected-utility model of unemployment insurance, if higher UI benefits do not change job-search behavior or unemployment durations, full insurance between employment and unemployment states is efficient.

- (c) Suppose that Obamacare had done only one thing: made it illegal for health insurance providers to deny applicants based on preexisting conditions (“community rating”). This policy would have reduced adverse selection.

- (d) Evidence that more generous unemployment insurance increases unemployment durations implies that the optimal unemployment insurance benefit must be zero.

- (e) If you believe that the moral hazard cost of disability insurance identified by judges-design evidence is larger than its social benefit for marginal applicants, then you should want to eliminate the disability insurance program altogether.

- (f) Evidence that reported workers' compensation claims for hard-to-verify back injuries spike on Mondays is consistent with moral hazard.

2. Empirical Problem: High Point University and the Top 20% [30 points]

We have uploaded a PDF of the Wall Street Journal article “How a Small North Carolina College Became a Magnet for Wealthy Students” with the problem set materials; the article is also available here. For the data, use the Opportunity Insights repository here and the Chetty-Friedman-Saez-Turner-Yagan (2020) college mobility paper here.

We want to determine whether the January 2005 arrival of High Point president Nido Qubein changed the share of High Point students with top-20%-income parents. In the longitudinal file, `cohort` is birth cohort, so age-20 year is `cohort+20`, and `par_q5` is the share of students with top-20%-income parents. Because students first admitted under Qubein in fall 2005 would typically appear in the age-20 data in 2007, treat 2007 as the first affected age-20 cohort, draw the vertical line in your event-study graph at 2006.5, let $HIGHPOINT_c = 1$ if college c is High Point University and $HIGHPOINT_c = 0$ otherwise, and let $POST_t = 1$ for $t \geq 2007$ and $POST_t = 0$ otherwise.

As a baseline control group, use colleges in Virginia, South Carolina, Georgia, and Tennessee that are private, non-profit, four-year institutions and classified as “Selective private” (tier 6) in the Opportunity Insights college-characteristics file, the same tier as High Point. This is a reasonable but imperfect starting point because it compares High Point to nearby schools in the same broad market segment, so you should assess it with the graph.

Use one observation per college-year and standard (i.e., unweighted ordinary least squares) regressions throughout. Do not worry about small-sample statistical inference issues like large standard errors; focus on the design, the graph, and the coefficient estimates. You may complete the empirical work in a notebook using <https://datahub.berkeley.edu/>, or use AI tools. In your submitted solution file, include screenshots of the relevant notebook output (graph and regression results) alongside your written answers.

(a) [4 points] From the Opportunity Insights data repository, download the following three files under “Mobility Report Cards”:

- i. “Preferred Estimates of Access and Mobility Rates by College” (Table 1)
- ii. “Baseline Longitudinal Estimates of Child and Parent Income Distributions by College and Child’s Cohort” (Table 3)
- iii. “College Level Characteristics from the IPEDS Database and the College Scorecard” (Table 10)

Identify High Point University in these files. Report High Point’s `super_opeid`, `state`, `sector`, and `tier`.

- (b) [5 points] Using the college-characteristics file, construct the baseline control group described above. How many colleges are in your control group? Are the control-group colleges relatively evenly divided across the four states, or are they mostly concentrated in a single state?

- (c) [7 points] Using the longitudinal file, create a graph with the share of students with top-20%-income parents on the y-axis and year on the x-axis. Use `par_q5` for the y-variable and `cohort+20` for the x-variable, where the x-axis year denotes the year that the students turn age 20. Plot two lines: one line for High Point, and a second line that is the simple unweighted mean value for the control-group colleges. Draw a vertical line at 2006.5. Do the treatment and control groups appear to exhibit parallel trends before 2007?

(d) [6 points] Estimate the overall difference-in-differences specification

$$y_{ct} = \alpha_c + \gamma_t + \delta (HIGHPOINT_c \times POST_t) + \varepsilon_{ct},$$

where y_{ct} is the top-20% parent share for college c in age-20 year t , α_c are college fixed effects, and γ_t are year fixed effects. Report and interpret $\hat{\delta}$.

- (e) [4 points] Estimate a “2011 DD” effect by restricting the sample to the pre period ($t < 2007$) and 2011, and then redefining $POST_t = 1$ for $t = 2011$ and $POST_t = 0$ for $t < 2007$. Run

$$y_{ct} = \alpha_c + \beta POST_t + \delta (HIGHPOINT_c \times POST_t) + \varepsilon_{ct}.$$

Report and interpret $\hat{\delta}$.

- (f) [4 points] Based on these empirical results and on what we discussed in class about how financial aid works, what might President Qubein's long-run financial strategy have been? There is no right answer here, but give a reasonable speculation. If it helps to have an example, think about the Wall Street Journal article that touted very expensive amenities like the planetarium, which was built in 2018 and cost millions of dollars.

3. Analytical Question: Stylized Unemployment Insurance [20 points]

A government is designing an unemployment insurance program. The individual earns wage $w = \$10$ if employed and earns \$0 if unemployed. The government offers benefit b if unemployed, financed by a tax p levied regardless of employment status. Let s denote the probability that the individual is unemployed and claims benefits; she can choose $s \in [0.01, 0.99]$.

Crucially, the individual can influence s through her own effort, such as searching harder for work. This is the source of *moral hazard*: the generosity of unemployment insurance may itself affect behavior. The individual's expected utility is:

$$E[U] = (1 - s) \ln(w - p) + s \ln(b - p) - (1 - s)^2$$

Here $w - p$ is consumption if employed, $b - p$ is consumption if unemployed, and the final term $-(1 - s)^2$ captures the increasing personal cost of reducing s .

Note: Parts (g) through (k) can be completed successfully without using the answers from parts (a) through (f). For questions that depend on answers to previous parts, you can still get full credit if your logic is correct even if your answer to a previous part is wrong. Therefore, be sure to explicitly state your logic.

- (a) [2 points] The government first considers Program #1, which provides *full insurance*—equalizing consumption across employment and unemployment. Suppose the government believes the probability of unemployment is 10%. What benefit b and tax p make this program budget balanced? *Hint:* expected tax revenue is p , and expected benefit payments are sb .

- (b) [2 points] Write out the individual's optimization problem under Program #1. Solve for her optimal choice of s . How does this compare to the government's assumed unemployment probability of 10%?

(c) [1 point] What is the individual's expected utility under Program #1?

- (d) [2 points] Suppose the government keeps the tax at $p = 1$ from Program #1 even after the individual's behavioral response from part (b). Does Program #1 run a surplus, break even, or run a deficit? Explain why this is an example of moral hazard rather than adverse selection.

- (e) [1 point] Based on the social-insurance logic from lecture, does the fiscal imbalance in part (d) suggest that full insurance is likely to be optimal in this setting? Briefly explain the intuition.

- (f) [2 points] Suppose the government nevertheless insists on providing full insurance, but adjusts the tax to keep the program budget balanced after accounting for the individual's behavioral response from part (b). What tax is required? What is the individual's expected utility under this budget-balanced full-insurance program? Call this **Program #2**.

Now suppose that, in light of the moral hazard problem identified above, the government moves away from full insurance. It instead offers a *partial*-insurance program with $b = 3$. The government expects that this lower benefit will induce the individual to choose $s = \frac{1}{3}$. Call this **Program #3**.

- (g) [2 points] What tax p makes Program #3 budget balanced, given the government's expectation that $s = \frac{1}{3}$?

- (h) [2 points] Verify the government's expectation: solve for the individual's optimal s under Program #3. Does the individual indeed choose $s = \frac{1}{3}$?

(i) [2 points] What is the individual's utility under Program #3?

- (j) [2 points] Given the individual's actual choice of s from part (h), does Program #3 run a surplus, break even, or run a deficit? Explain briefly.

- (k) [2 points] Compare Programs #2 and #3 from a social welfare perspective. Which program should the government prefer, and why? Your answer should explicitly discuss the trade-off between *consumption smoothing* (the benefit of insurance) and *moral hazard* (the cost of insurance), and draw a general lesson for the design of unemployment insurance. **Note:** If you did not successfully compute utility under Program #2, you may use the fact that it is less than 0.